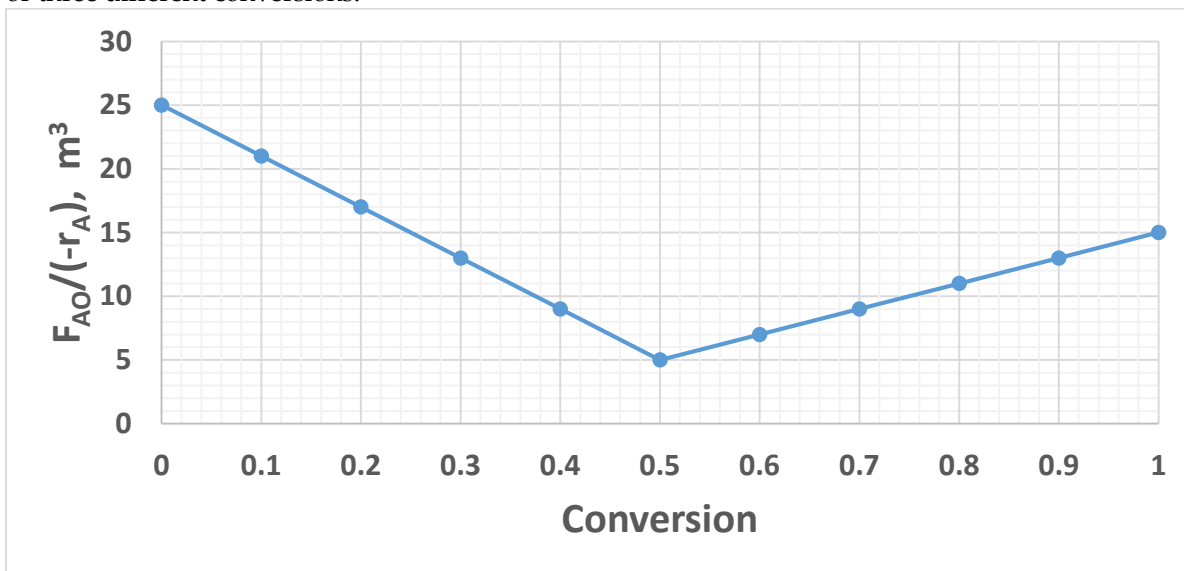
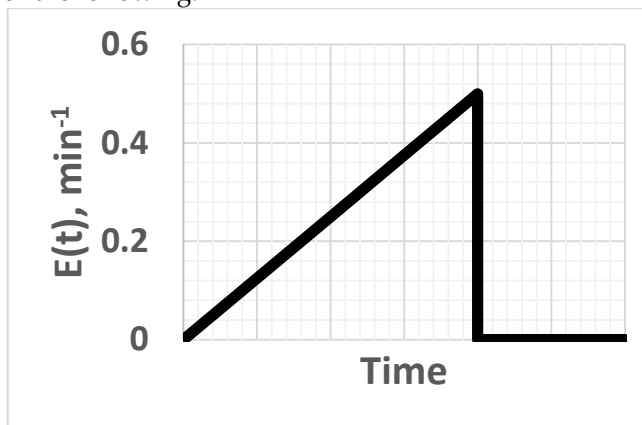


CHE 321 Chemical Reaction Engineering, FINAL EXAM, 100 points
Due on 6 May at 1:30 PM

1. (10 points) The following Levenspiel plot represents a liquid phase reaction under adiabatic conditions for one component (A). Clearly, the same volume can provide multiple conversions (X_1 , X_2 and/or X_3) at different temperatures if the reaction is conducted in a CSTR. Find the equation that relates the two or three different conversions.



2. (4 points) What experiments will you propose to determine the mass-transfer or pore-diffusion limitations in a catalytic reactor?
3. (4 points) Find the $E(t)$ for the following:



4. (20 points) An elementary irreversible liquid-phase reaction $A + B \rightarrow C$ is carried out in a CSTR. A and B are fed at molar rates of 1.25 mol/s and 1 mol/s respectively, at a temperature of 300 K. The reactor is jacketed and the jacket temperature can be assumed to be 310 K. An agitator contributes a work of 20.9 kW to the reactor. The volumetric flow rate is 5 lit/s. Additionally:

$H_A^0(298K) = -20 \text{ kcal/mol}$; $H_B^0(298 K) = -25 \text{ kcal/mol}$; $H_C^0(298 K) = -60 \text{ kcal/mol}$

$C_{PA} = C_{PB} = 40 \text{ cal/mol/K}$; $C_{PC} = 55 \text{ cal/mol/K}$

$k = 0.01 \text{ Lit/mol/s}$ at $300K$; $E_A = 8 \text{ kcal/mol}$

$UA = 75 \text{ Cal/s/K}$

Determine the volume of the reactor for 60% conversion of A.

5. (35 points) A reaction that is $\frac{1}{2}$ order is being carried out in a reactor that can be modeled as three PFRs in parallel with residence times of 30 sec, 50 sec and 75 sec and flow divided in 1:2:3 ratio. The $C_{A0} = 1 \text{ mol/lit}$ and the rate constant is $0.25 \text{ mol}^{1/2}/\text{lit}^{1/2}/\text{sec}$.
- Find the residence time distribution function for the reactor
 - What is the average residence time for the reactor?
 - Determine the conversion for the above reaction in this reactor using the segregated flow model.
 - Do you expect the conversion to be more or less for maximum mixedness model?
6. (27 points) The kinetic data for a 2nd order gas-phase reaction was obtained in a fluidized bed reactor with excessive mixing for different particle sizes. The results are shown in the table below.

Particle Size (mm)	Rate of Reaction: $-r_A$ (mole/cm ³ /s)	Bulk Concentration C_{Ab} (mole/cm ³)
0.01	8.9	0.01
0.02	8.9	0.01
0.05	8.9	0.01
0.07	8.8	0.02
0.09	5.55	0.04
0.11	4.06	0.05
0.3	1.49	0.05
0.5	0.89	
1		0.05

The diffusivity of the fluid is $D_e = 1 \times 10^{-4} \text{ cm}^2/\text{s}$.

- What is the rate constant for the reaction?
- For the above table, what is the bulk-concentration for particle size of 0.5 mm; and what is the reaction rate for 1 mm?
- Find the Thiele Modulus and the effectiveness factor for all the particle sizes.